Ex. No: 2

CREATE TABLE Student (StudentID int, Name varchar(30), Age int);

Output:

Table created.

INSERT INTO Student VALUES (1, 'Alice', 20);
INSERT INTO Student VALUES (2, 'Bob', 22);
INSERT INTO Student VALUES (3, 'Charlie', 21);

Output:

3 rows created.

CREATE TABLE Courses (CourseID int, CourseName varchar(20));

Output:

Table created.

INSERT INTO Courses VALUES (101, 'Math');
INSERT INTO Courses VALUES (102, 'English');
INSERT INTO Courses VALUES (103, 'History');

Output:

3 rows created.

CREATE TABLE Enrollments (EnrollmentID int, StudentID int, CourseID int, Grade varchar(5));

Output:

Table created.

INSERT INTO Enrollments VALUES (1, 1, 101, 'A');
INSERT INTO Enrollments VALUES (2, 1, 102, 'B');
INSERT INTO Enrollments VALUES (3, 2, 101, 'A-');
INSERT INTO Enrollments VALUES (4, 3, 103, 'B+');
INSERT INTO Enrollments VALUES (5, 3, 102, 'A');

Output:

5 rows created.

INNER JOIN:

SELECT Student.StudentID, Student.Name, Student.Age, Courses.CourseID, Courses.CourseName, Enrollments.Grade FROM Student INNER JOIN Enrollments ON Student.StudentID = Enrollments.StudentID INNER JOIN Courses ON Enrollments.CourseID = Courses.CourseID;

Output:

StudentID	Name	Age	CourseID	CourseName	Grade
1	Alice	20	101	Math	A
1	Alice	20	102	English	B
2	Bob	22	101	Math	A-
3	Charlie	21	103	History	B+
3	Charlie	21	102	English	A

LEFT JOIN:

SELECT Student.StudentID, Student.Name, Student.Age, Courses.CourseID, Courses.CourseName, Enrollments.Grade FROM Student LEFT JOIN Enrollments ON Student.StudentID = Enrollments.StudentID LEFT JOIN Courses ON Enrollments.CourseID = Courses.CourseID;

Output:

StudentID	Name	Age	CourseID	CourseName	Grade
1	Alice	20	101	Math	A
1	Alice	20	102	English	B
2	Bob	22	101	Math	A-
3	Charlie	21	103	History	B+
3	Charlie	21	102	English	A

RIGHT JOIN:

SELECT Student.StudentID, Student.Name, Student.Age, Courses.CourseID, Courses.CourseName, Enrollments.Grade FROM Courses RIGHT JOIN Enrollments ON Courses.CourseID = Enrollments.CourseID RIGHT JOIN Student ON Enrollments.StudentID = Student.StudentID;

Output:

StudentID	Name	Age	CourseID	CourseName	Grade
1	Alice	20	101	Math	A
1	Alice	20	102	English	B
3	Charlie	21	103	History	B+
2	Bob	22	101	Math	A-
3	Charlie	21	102	English	A